

Curve Sketching

Example 1: Sketch and properly labelled graph of $f(x) = \frac{x}{(3x+1)^2}$, given

$$f'(x) = \frac{-3x+1}{(3x+1)^3} \text{ and } f''(x) = \frac{6(3x-2)}{(3x+1)^4}.$$

* Domain: $(-\infty, -\frac{1}{3}) \cup (-\frac{1}{3}, \infty)$

* NO HOLES

* V.A. @ $x = -\frac{1}{3}$

* x & y-int

→ x-int: set $y=0$

$$\Rightarrow x=0$$

$$\therefore \text{x-int @ } (0,0)$$

→ y-int: set $x=0$

$$\Rightarrow y=0$$

$$\therefore \text{y-int @ } (0,0)$$

* Behaviour Near V.A.

$$\lim_{x \rightarrow -\frac{1}{3}^-} \frac{x}{(3x+1)^2} = -\infty$$

$$\lim_{x \rightarrow -\frac{1}{3}^+} \frac{x}{(3x+1)^2} = -\infty$$

* H.A.

$$\text{H.A. @ } y=0$$

→ $f(100) = \frac{100}{(301)^2} > 0 \Rightarrow$ above H.A. as $x \rightarrow \infty$

→ $f(-100) = \frac{-100}{(-299)^2} < 0 \Rightarrow$ below H.A. as $x \rightarrow -\infty$

H.A. Cross test

$$\frac{x}{(3x+1)^2} = 0 \Rightarrow x=0 \therefore \text{crosses H.A. @ } (0,0)$$

* Local Extrema

critical # is $x = \frac{1}{3}$

* $x = -\frac{1}{3}$ (V.A.)

Interval $(-\infty, -\frac{1}{3})$ $(-\frac{1}{3}, \frac{1}{3})$ $(\frac{1}{3}, \infty)$

$-3x+1$ + + -

$(3x+1)^3$ - + +

$f'(x)$ - + -

$f(x)$ ↓ ↑ ↓

↑
V.A. @
 $x = -\frac{1}{3}$

↑
Max @ $x = \frac{1}{3}$ pt. $(\frac{1}{3}, \frac{1}{12})$

* Concavity & Possible Poi

$f''(\frac{2}{3}) = 0$; $f''(-\frac{1}{3}) \text{ DNE (V.A.)}$

$x = \frac{2}{3}$ and $x = -\frac{1}{3}$ (V.A.)

Interval $(-\infty, -\frac{1}{3})$ $(-\frac{1}{3}, \frac{2}{3})$ $(\frac{2}{3}, \infty)$

$3x-2$ - - +

$(3x+1)^4$ + + +

$f''(x)$ - - +

$f(x)$ c.D. c.D. c.U.

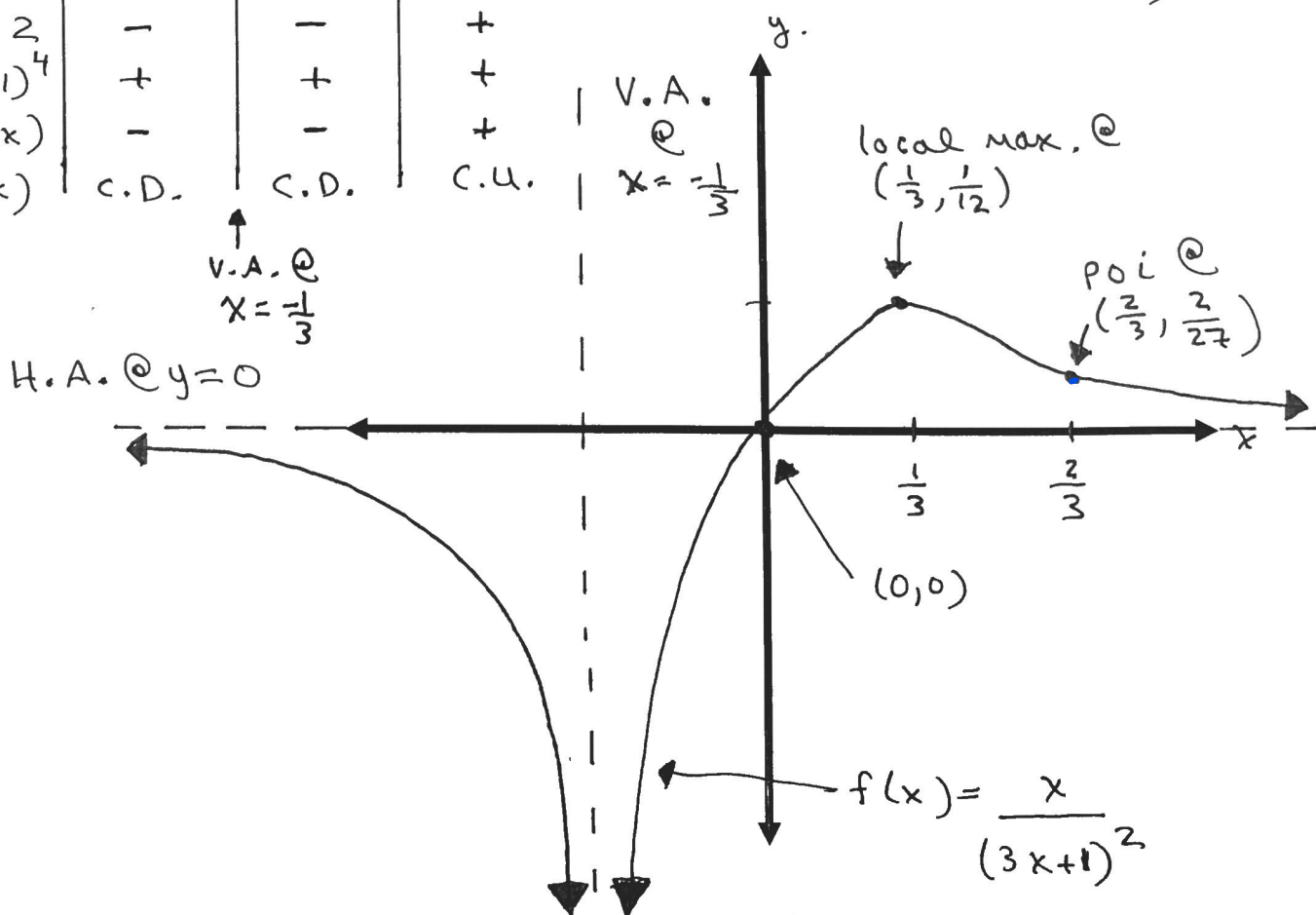
V.A.
@
 $x = -\frac{1}{3}$

$\therefore$ poi @ $(\frac{2}{3}, \frac{2}{27})$

local max. @
 $(\frac{1}{3}, \frac{1}{12})$

poi @
 $(\frac{2}{3}, \frac{2}{27})$

H.A. @ $y=0$



Example 2: Sketch and properly labelled graph of $f(x) = -3x^{\frac{5}{3}} + 15x^{\frac{2}{3}}$, given

$$f'(x) = -5x^{-\frac{1}{3}}(x-2) \text{ and } f''(x) = -\frac{10}{3}x^{-\frac{4}{3}}(x+1).$$

$$= -3x^{\frac{2}{3}}(x-5)$$

$\uparrow$ $\uparrow$
 $x=0$ $x=5$

* Domain: $(-\infty, \infty)$

* NO HOLES

* NO V.A.

* NO H.A. / NO O.A.

* x & y-int

→ x-int: set $y=0$

$$\Rightarrow x=0, 5$$

∴ x-int @ $(0,0)$ & $(5,0)$

→ y-int: set $x=0$

$$\Rightarrow y=0$$

∴ y-int @ $(0,0)$

* Local Extrema

critical #'s are $x=0, 2$

Interval	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
-5	-	-	-
$x^{\frac{1}{3}}$	-	+	+
$x-2$	-	-	+
$f'(x)$	-	+	-
$f(x)$	↓	↑	↓

Min @ $(0,0)$

Max @ $(2, 14.29)$

* Concavity & possible poi

$$f''(0) \text{ DNE} ; f''(-1) = 0$$

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, \infty)$
$-\frac{10}{3}$	-	-	-
$x^{4/3}$	+	+	+
$x+1$	-	+	+
$f'''(x)$	+	-	-
$f(x)$	C.U.	C.D.	C.D.

poi @ $(-1, 18)$

No poi @ $(0, 0)$ "cusp"

$$f(x) = -3x^{\frac{5}{3}} + 15x^{\frac{2}{3}}$$

